

LISTA DE EXERCÍCIOS FINAL

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Problema 1 - Visão Geral de um Sistema Computacional (*Desktop* ou *Notebook*)

a)

Processador: AMD Ryzen 5 5600;
Tamanho da palavra: 64 bits;
Frequência do Clock: 4470 Mhz;
Número de núcleos: 6;
Monothread ou Multithread: *multithread*.

```
$ lscpu [19:01:00]
Arquitetura: x86_64
Modo(s) operacional da CPU: 32-bit, 64-bit
Tamanhos de endereço: 48 bits physical, 48 bits virtual
Ordem dos bytes: Little Endian
CPU(s): 12
Lista de CPU(s) on-line: 0-11
ID de fornecedor: AuthenticAMD
Nome do modelo: AMD Ryzen 5 5600 6-Core Processor
Família da CPU: 25
Modelo: 33
Thread(s) per núcleo: 2
Núcleo(s) por soquete: 6
Soquete(s): 1
Step: 2
Aumento de frequência: habilitado
CPU(s) MHz de escala: 77%
CPU MHz máx.: 4470,5078
CPU MHz min.: 565,5470
BogoMIPS: 6986,45
Opções: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mc
a cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall n
x mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_go
od nopl xtopology nonstop_tsc cpuid extd_apicid aperfmp
erf rapl pni pclmulqdq monitor ssse3 fma cx16 sse4_1 ss
e4_2 x2apic movbe popcnt aes xsave avx f16c rdrand lahf
_lm cmp_legacy svm extapic cr8_legacy abm sse4a misalig
nsse 3dnowprefetch osvw ibs skinit wdt tce topoext perf
ctr_core perfctr_nb bpext perfctr_llc mwaitx cpb cat_l3
cdp_l3 hw_pstate ssbd mba ibrs ibpb stibp vmmcall fsgs
base bmi1 avx2 smep bmi2 erms invpcid cqm rdt_a rdseed
adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1
xsaves cqm_llc cqm_occup_llc cqm_mbm_total cqm_mbm_loc
al user_shstk clzero irperf xsaveerptr rdpru wbnoinvd a
rat npt lbrv svm_lock nrip_save tsc_scale vmcb_clean fl
ushbyasid decodeassists pausefilter pfthreshold avic v_
vmsave_vmload vgif v_spec_ctrl umip pku ospke vaes vpc_
mulqdq rdpid overflow_recov succor smca fsrm debug_swap

Recursos de virtualização:
```

b)

Tipo: DDR4;
Capacidade: 16GB;
Frequência de Clock: 2400 Mhz.

```

Handle 0x0015, DMI type 17, 92 bytes
Memory Device
    Array Handle: 0x000B
    Error Information Handle: 0x0014
    Total Width: 64 bits
    Data Width: 64 bits
    Size: 8 GB
    Form Factor: DIMM
    Set: None
    Locator: DIMM 1
    Bank Locator: P0 CHANNEL A
    Type: DDR4
    Type Detail: Synchronous Unbuffered (Unregistered)
    Speed: 2400 MT/s
    Manufacturer: Unknown
    Serial Number: 0000046C
    Asset Tag: Not Specified
    Part Number: VMA47UG-MEC1U22T3
    Rank: 1
    Configured Memory Speed: 2400 MT/s
    Minimum Voltage: 1.2 V
    Maximum Voltage: 1.2 V
    Configured Voltage: 1.2 V
    Memory Technology: DRAM
    Memory Operating Mode Capability: Volatile memory
    Firmware Version: Unknown
    Module Manufacturer ID: Bank 9, Hex 0xF1
    Module Product ID: Unknown
    Memory Subsystem Controller Manufacturer ID: Unknown
    Memory Subsystem Controller Product ID: Unknown
    Non-Volatile Size: None
    Volatile Size: 8 GB
    Cache Size: None
    Logical Size: None

```

Esse *print* corresponde apenas a um dos pentes de memória. Mas existem 2 (desse modelo) na máquina em questão

c)

Cache L1d (cache L1 de dados): 192 KiB;
Cache L1i (cache L1 de instruções): 192 KiB;
Cache L2: 3 MiB;
Cache L3: 32 MiB.

```

$ lscpu | grep -i cache [19:11:31]
cache de L1d:           192 KiB (6 instâncias)
cache de L1i:           192 KiB (6 instâncias)
cache de L2:            3 MiB (6 instâncias)
cache de L3:            32 MiB (1 instância)

```

d)

Tipo: SSD (Rota = 0);
Capacidade: aprox. 1 TB;
Tipo do barramento: NVMe.

```
$ lsblk -o NAME,SIZE,TYPE,ROTA,TRAN [19:13:03]
NAME          SIZE TYPE  ROTA TRAN
zram0         15,5G disk    0
nvme0n1       931,5G disk    0 nvme
├─nvme0n1p1    100M part    0 nvme
├─nvme0n1p2     16M part    0 nvme
├─nvme0n1p3   832,4G part    0 nvme
├─nvme0n1p4    720M part    0 nvme
├─nvme0n1p5    477M part    0 nvme
├─nvme0n1p6     96M part    0 nvme
├─nvme0n1p7   1024M part    0 nvme
├─nvme0n1p8    92,1G part    0 nvme
├─nvme0n1p9     4G part    0 nvme
└─cryptswap    4G crypt    0
```

e)

Tamanho da memória virtual: 20 GB.

```
$ free -h [19:24:05]
              total        usada        livre      compart.  buff/cache  disponível
Mem.:         15Gi         5,6Gi         8,4Gi         109Mi         2,7Gi         9Gi
Swap:         19Gi           0B         19Gi
```

f)

Mouse

Modo de transmissão: Serial (pois está conectado via USB);

Tipo de transmissão: Assíncrona, o que pode ser percebido através da nomenclatura *Interrupt*, que indica que o dispositivo “interrompe” o processador para enviar dados.

Bus 003 Device 002: ID 046d:c092 Logitech, Inc. G102/G203 LIGHTSYNC Gaming Mouse
Couldn't open device, some information will be missing

Device Descriptor:

bLength 18
bDescriptorType 1
bcdUSB 2.00
bDeviceClass 0 [unknown]
bDeviceSubClass 0 [unknown]
bDeviceProtocol 0
bMaxPacketSize0 64
idVendor 0x046d Logitech, Inc.
idProduct 0xc092 G102/G203 LIGHTSYNC Gaming Mouse
bcdDevice 52.00
iManufacturer 1 Logitech
iProduct 2 G203 LIGHTSYNC Gaming Mouse
iSerial 3 206736764D47
bNumConfigurations 1

Configuration Descriptor:

bLength 9
bDescriptorType 2
wTotalLength 0x003b
bNumInterfaces 2
bConfigurationValue 1
iConfiguration 4
bmAttributes 0xa0
(Bus Powered)
Remote Wakeup
MaxPower 300mA

Interface Descriptor:

bLength 9
bDescriptorType 4
bInterfaceNumber 0
bAlternateSetting 0
bNumEndpoints 1
bInterfaceClass 3 Human Interface Device
bInterfaceSubClass 1 Boot Interface Subclass
bInterfaceProtocol 2 Mouse
iInterface 0

HID Device Descriptor:

bLength 9
bDescriptorType 33
bcdHID 1.11
bCountryCode 0 Not supported
bNumDescriptors 1
bDescriptorType 34 Report

```

wDescriptorLength      67
Report Descriptors:
  ** UNAVAILABLE **
Endpoint Descriptor:
  bLength              7
  bDescriptorType      5
  bEndpointAddress     0x81  EP 1 IN
  bmAttributes         3
    Transfer Type      Interrupt
    Synch Type         None
    Usage Type         Data
  wMaxPacketSize       0x0008  1x 8 bytes
  bInterval            1
Interface Descriptor:
  bLength              9
  bDescriptorType      4
  bInterfaceNumber     1
  bAlternateSetting     0
  bNumEndpoints        1
  bInterfaceClass      3 Human Interface Device
  bInterfaceSubClass   0 [unknown]
  bInterfaceProtocol   0
  iInterface           0
  HID Device Descriptor:
    bLength            9
    bDescriptorType    33
    bcdHID             1.11
    bCountryCode       0 Not supported
    bNumDescriptors    1
    bDescriptorType    34 Report
    wDescriptorLength  151
    Report Descriptors:
      ** UNAVAILABLE **
Endpoint Descriptor:
  bLength              7
  bDescriptorType      5
  bEndpointAddress     0x82  EP 2 IN
  bmAttributes         3
    Transfer Type      Interrupt
    Synch Type         None
    Usage Type         Data
  wMaxPacketSize       0x0014  1x 20 bytes
  bInterval            1

```

Teclado

Modo de transmissão: Serial;

Tipo de transmissão: Assíncrona, pelos mesmos motivos do mouse.

Bus 001 Device 002: ID 3554:fa09 CompX 2.4G Wireless Receiver
Couldn't open device, some information will be missing

Device Descriptor:

bLength 18
bDescriptorType 1
bcdUSB 2.00
bDeviceClass 0 [unknown]
bDeviceSubClass 0 [unknown]
bDeviceProtocol 0
bMaxPacketSize0 64
idVendor 0x3554 CompX
idProduct 0xfa09 2.4G Wireless Receiver
bcdDevice 30.00
iManufacturer 1 CompX
iProduct 2 2.4G Wireless Receiver
iSerial 0
bNumConfigurations 1

Configuration Descriptor:

bLength 9
bDescriptorType 2
wTotalLength 0x003b
bNumInterfaces 2
bConfigurationValue 1
iConfiguration 0
bmAttributes 0xa0
(Bus Powered)
Remote Wakeup
MaxPower 98mA

Interface Descriptor:

bLength 9
bDescriptorType 4
bInterfaceNumber 0
bAlternateSetting 0
bNumEndpoints 1
bInterfaceClass 3 Human Interface Device
bInterfaceSubClass 1 Boot Interface Subclass
bInterfaceProtocol 1 Keyboard
iInterface 0

HID Device Descriptor:

bLength 9
bDescriptorType 33
bcdHID 1.11
bCountryCode 0 Not supported
bNumDescriptors 1


```

    bDescriptorType      34 Report
    wDescriptorLength    77
    Report Descriptors:
      ** UNAVAILABLE **
Endpoint Descriptor:
  bLength                7
  bDescriptorType        5
  bEndpointAddress      0x81  EP 1 IN
  bmAttributes          3
    Transfer Type        Interrupt
    Synch Type           None
    Usage Type           Data
  wMaxPacketSize        0x0008  1x 8 bytes
  bInterval              1
Interface Descriptor:
  bLength                9
  bDescriptorType        4
  bInterfaceNumber      1
  bAlternateSetting      0
  bNumEndpoints          1
  bInterfaceClass        3 Human Interface Device
  bInterfaceSubClass     1 Boot Interface Subclass
  bInterfaceProtocol     2 Mouse
  iInterface             0
  HID Device Descriptor:
    bLength              9
    bDescriptorType      33
    bcdHID               1.11
    bCountryCode         0 Not supported
    bNumDescriptors      1
    bDescriptorType      34 Report
    wDescriptorLength    218
    Report Descriptors:
      ** UNAVAILABLE **
Endpoint Descriptor:
  bLength                7
  bDescriptorType        5
  bEndpointAddress      0x82  EP 2 IN
  bmAttributes          3
    Transfer Type        Interrupt
    Synch Type           None
    Usage Type           Data
  wMaxPacketSize        0x0019  1x 25 bytes
  bInterval              1

```

Problema 2 - Visão Geral de um Sistema Computacional (*Smartphone*)

a)

Processador: T8101 Apple A14 Bionic;
Tamanho da palavra: 64 bits;
Frequência do *Clock*: até 3,1 GHz;
Número de núcleos: 2.

Chipset

T8101 Apple A14 Bionic

64 Bit



-
- **2 núcleos de alto desempenho Firestorm** com clock de até 3,1 GHz para tarefas exigentes;
 - **4 núcleos de alta eficiência Icestorm** com clock em torno de 1,8 GHz para processos em segundo plano.

b)

Capacidade: 6 GB.

c)

Capacidade: 512 GB.

RAM	6 GB
Memória Max	512 GB

d) Existente, mas reservada a processos internos do kernel. Não encontrei a informação.i

e)

Denominação: Apple GPU (4-core)

GPU	Apple GPU (4-core)
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